CLINICAL · APPROACH TO

# Approach to a sore throat

A sore throat is one of the commonest reasons anyone sees a doctor, and almost all acute sore throats are viral and self-limiting. The ENT surgeon sees the other end of the spectrum: the patient who cannot swallow saliva, the quinsy, the supraglottitis that will obstruct the airway, the neck space abscess, and the adult with a persistent one-sided sore throat and earache that is a cancer. The approach separates acute from chronic, recognises the airway and the deep collection, uses antibiotics with judgement, and knows when tonsillectomy is justified.

AT A GLANCE

- Centor or FeverPAIN to decide on antibiotics.
- Trismus, drooling, muffled voice: quinsy or deep neck abscess.
- Stridor or toxic patient: epiglottitis — airway first.
- Over 3 weeks, unilateral, otalgia: examine for cancer.

## 1 CHAPTER ONE Acute or chronic?

Then: is the airway safe?

### 1.1 The first questions

An **acute** sore throat lasts days to two weeks and is usually infective. A **chronic** sore throat persists beyond three weeks and has a different list: reflux, chronic tonsillitis, tonsilloliths, postnasal drip, smoking, voice misuse, and malignancy. In either case the first judgement is about the airway and the swallow.

RED FLAGS IN SORE THROAT

- **Drooling, inability to swallow saliva, muffled voice, stridor or sitting forward:** supraglottitis or a deep neck abscess — protect the airway, do not lie the patient flat, do not examine the throat with a tongue depressor in a child.
- **Trismus with unilateral swelling and uvular deviation:** quinsy.
- **Neck swelling, torticollis or reduced neck movement, especially in a child:** retropharyngeal or parapharyngeal abscess.

- **Persistent sore throat with fever, rigors and neck tenderness after tonsillitis:** Lemierre's syndrome.
- **Sore throat in a patient taking carbimazole, clozapine or chemotherapy:** neutropenia — urgent full blood count.
- **Persistent unilateral sore throat, referred otalgia, dysphagia, hoarseness, weight loss or a neck lump, especially in a smoker:** cancer until proven otherwise.

CHAPTER TWO

# 2 The acute sore throat

Assess, score, decide.

## 2.1 History and examination

Ask about the duration, fever, ability to swallow food and fluids, voice change, breathing, earache and trismus, and about cough and coryza, which point to a viral cause. Examine the oropharynx, the tonsils, the neck nodes and the airway, and, when the history suggests supraglottitis, examine the larynx with a flexible endoscope in a safe setting rather than with a tongue depressor.

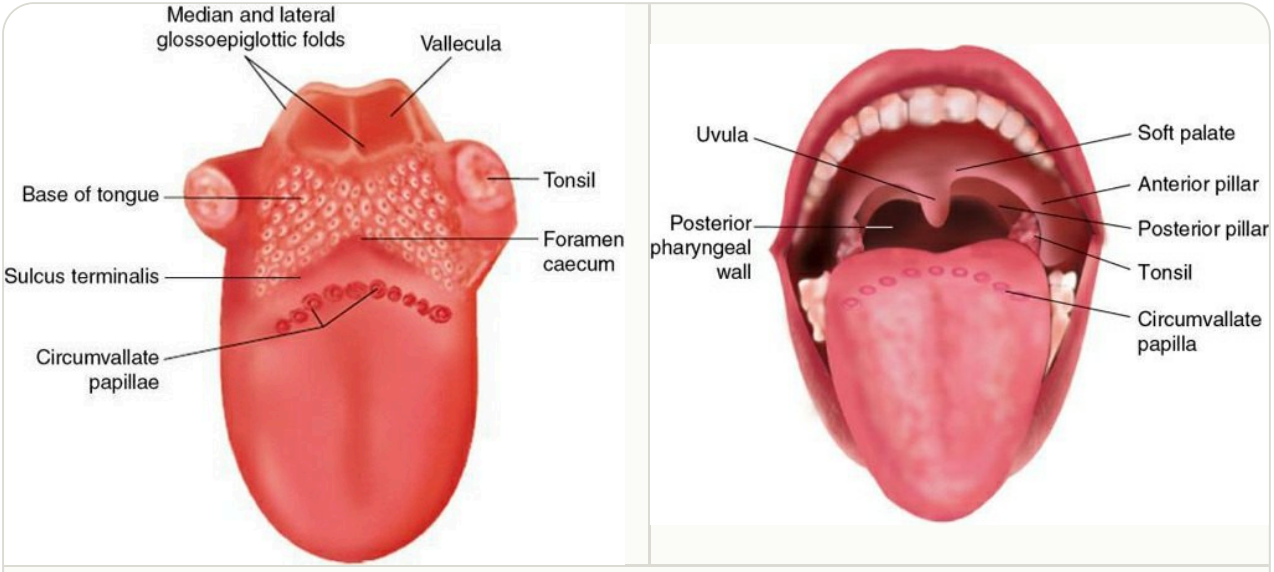



**FIGURE 1** First, the base of the tongue and valleculae. Second, the structures seen in the oropharynx.

Dhingra, figs 47.6 and 47.7

## 2.2 Who needs antibiotics?

Most acute sore throats are viral, and antibiotics shorten symptoms by only about a day in bacterial tonsillitis. Scoring systems identify patients more likely to have streptococcal infection. **FeverPAIN** scores one point each for fever, purulence, attending rapidly (within three days), inflamed tonsils and no cough or coryza: a score of 0 to 1 needs no antibiotic, 2 to 3 a delayed or no antibiotic, and 4 to 5 an immediate antibiotic or a delayed one. The **Centor** criteria (tonsillar exudate, tender anterior cervical nodes, fever and absence of cough) are used similarly, with antibiotics considered at three or four. Phenoxymethylpenicillin for five to ten days is the first choice, with clarithromycin in penicillin allergy. **Amoxicillin is avoided** because it causes a rash in infectious mononucleosis.

Admission is needed for patients who cannot swallow fluids, are dehydrated, have a suspected collection or airway compromise, or are systemically unwell. They need intravenous fluids, analgesia, intravenous benzylpenicillin and often a single dose of dexamethasone.

## 2.3 The important acute causes

**Acute tonsillitis** is viral or bacterial (group A streptococcus is the commonest bacterium). **Infectious mononucleosis** from Epstein–Barr virus gives a severe exudative tonsillitis with palatal petechiae, generalised lymphadenopathy and splenomegaly; it is confirmed by a monospot or EBV serology and atypical lymphocytes. Patients avoid contact sports for several weeks because of splenic rupture, and steroids are used for airway compromise. **Peritonsillar abscess (quinsy)** produces trismus, a hot potato voice, unilateral swelling above the tonsil and a deviated uvula; it is drained by needle aspiration or incision. **Supraglottitis** produces severe odynophagia with relatively normal oropharynx, a muffled voice and drooling. **Deep neck space infections** need contrast CT and drainage. **Lemierre's syndrome** is a septic thrombophlebitis of the internal jugular vein, usually with *Fusobacterium necrophorum*, after a sore throat. Other causes include HIV seroconversion, diphtheria in the unvaccinated, gonococcal pharyngitis, and herpangina or hand, foot and mouth disease in children.

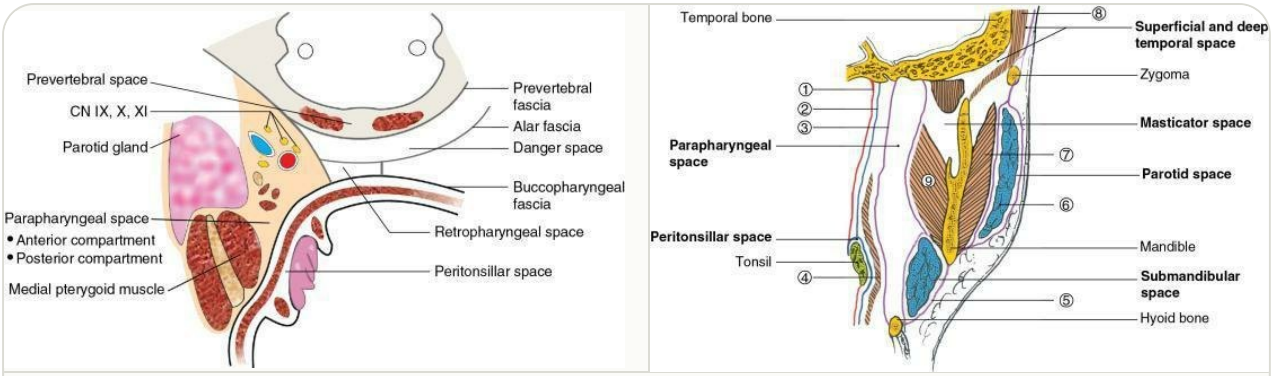



**FIGURE 2** First, the spaces around the pharynx where abscesses form. Second, the deep spaces of the head and neck.

Dhingra, figs 52.6 and 52.9

CHAPTER THREE

# 3 The chronic sore throat

A different list.

## 3.1 Causes

### CAUSES OF A CHRONIC SORE THROAT

| CAUSE | CLUES |
|---|---|
| Chronic or recurrent tonsillitis | Repeated episodes, halitosis, enlarged nodes |
| Tonsilloliths | Halitosis, white stones, foreign body sensation |
| Laryngopharyngeal reflux | Throat clearing, hoarseness, globus, morning symptoms |
| Postnasal drip | Nasal symptoms, throat clearing |
| Smoking, alcohol, dry mouth | History; mouth breathing; drugs |
| Voice misuse and muscle tension | Pain with voice use |
| Glossopharyngeal neuralgia and Eagle syndrome | Sharp pain triggered by swallowing; elongated styloid |
| Malignancy | Unilateral, progressive, otalgia, dysphagia, neck node, smoker |

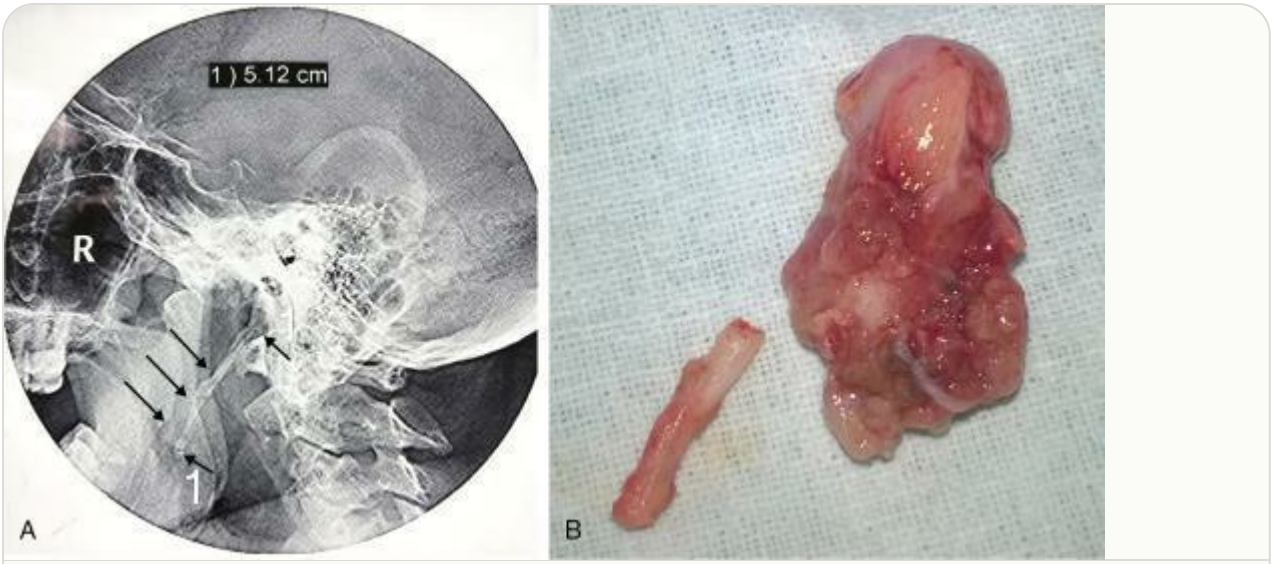



**FIGURE 3** An elongated styloid process (Eagle syndrome): (A) on radiograph, (B) excised.
Dhingra, fig. 53.6

### 3.2 Assessment

Every adult with a sore throat lasting more than three weeks, especially if unilateral, needs a full head and neck examination with flexible endoscopy of the nasopharynx, oropharynx, hypopharynx and larynx, palpation of the tongue base and tonsils, and examination of the neck. Asymmetrical tonsils in an adult, a firm or ulcerated tonsil, or a neck node lead to biopsy or tonsillectomy for histology, and imaging.

## 4 CHAPTER FOUR Tonsillectomy

When it is justified.

### 4.1 Indications

In adults and children, the **SIGN** criteria for tonsillectomy in recurrent tonsillitis are sore throats due to acute tonsillitis that are disabling and prevent normal functioning, with **seven or more** well-documented, clinically significant, adequately treated episodes in the preceding year, **five or more** in each of the preceding two years, or **three or more** in each of the preceding three years. Other indications

include two episodes of quinsy, suspected malignancy, and obstructive sleep-disordered breathing in children.

THE SORE THROAT

1 Airway and swallow

→ **drooling, stridor, muffled voice**
airway team, endoscopy in a safe setting, admit

→ **trismus with unilateral swelling**
quinsy — drain

→ **neck swelling or torticollis**
contrast CT for deep neck abscess

2 Acute, airway safe

→ **FeverPAIN 0–1**
no antibiotic

→ **2–3**
delayed or none

→ **4–5**
penicillin

→ **unable to swallow fluids**
admit, IV penicillin, fluids, steroid

→ **on carbimazole or clozapine**
urgent blood count

3 Glandular fever features?

→ monospot, avoid amoxicillin, no contact sport, steroids for airway compromise

4 Chronic, beyond three weeks

→ full endoscopic examination

→ **asymmetric tonsil, unilateral pain or neck node**
biopsy and imaging

→ **recurrent tonsillitis meeting criteria**
tonsillectomy

→ **reflux or postnasal drip**
treat the cause

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The airway red flags** and how to examine safely.
- **FeverPAIN and Centor;** antibiotic choice and avoiding amoxicillin.
- **Infectious mononucleosis:** diagnosis and advice.
- **Quinsy, deep neck infection and Lemierre's syndrome.**
- **SIGN criteria** for tonsillectomy.
- **The persistent unilateral sore throat** as a cancer symptom.

## 5.2 Common pitfalls

COMMON PITFALLS

- Examining a child with suspected supraglottitis with a tongue depressor.
- Giving amoxicillin in glandular fever.
- Missing agranulocytosis in a patient on carbimazole.
- Treating a unilateral chronic sore throat with repeated antibiotics in a smoker.

Written for this app from Oxford Textbook of Otolaryngology ch. 49, 76, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 50, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

## Tonsillar Abscess

### IMAGING FINDINGS

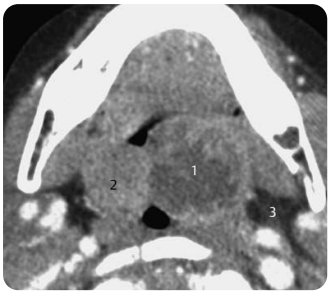
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

### TOP DIFFERENTIALS

- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

### CHECKLIST

- Airway
- Internal jugular vein thrombosis and carotid




CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

# Parapharyngeal Abscess

## IMAGING FINDINGS

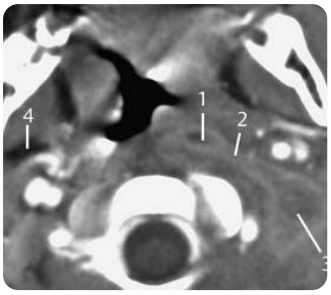
- Collection in the parapharyngeal fat, usually from the tonsil
- Medial bulge of the pharyngeal wall
- Trismus from medial pterygoid involvement

## TOP DIFFERENTIALS

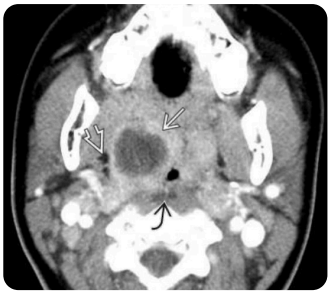
- Peritonsillar abscess
- Retropharyngeal abscess
- Deep lobe parotid infection

## CHECKLIST

- Carotid space and jugular vein (Lemierre)
- Spread to the mediastinum




CT with intravenous contrast enhancement, axial. Note the bulging left pharyngeal wall, as a result of a parapharyngeal abscess (1) which is…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

# Abscess of the Neck

## IMAGING FINDINGS

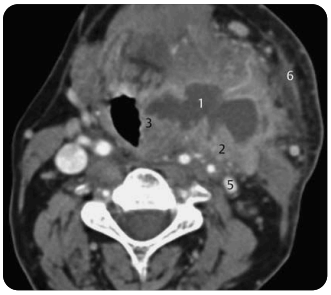
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

## TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

## CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route




CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

# Tonsillar Carcinoma

## IMAGING FINDINGS

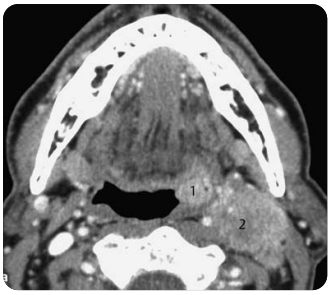
- Asymmetric enhancing tonsillar mass, sometimes small
- Spread to soft palate, tongue base and parapharyngeal space
- Cystic or necrotic level II nodes (HPV)

## TOP DIFFERENTIALS

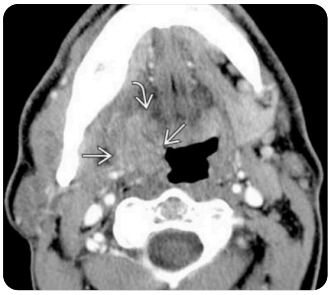
- Tonsillar lymphoid hyperplasia
- Lymphoma
- Peritonsillar abscess

## CHECKLIST

- Medial pterygoid, mandible and carotid
- Both sides of the neck




CT with intravenous contrast enhancement, axial. In the left tonsillar fossa, a small and enhancing tumor is visible (1), with enlarged level II lymph…



Axial CECT in alcoholic smoker presenting with multiple right neck masses shows an enlarged right palatine tonsil mass…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 492.